

SYMMETRY-COMPLETE INTEGRAL TRACE RECONSTRUCTION FOR THE PROJECTIVE LEECH LATTICE

THE CLANKERS: RESEARCH DRAFT

ABSTRACT. A nine-coordinate time readout of a projective Leech model fails to retain a second symmetry. We determine its smallest stable linear completion: all 759 Golay-octad traces, of rank 24. Their explicit inverse is $x_i = (23S_i - 7S)/1012$. Among integral, linearly compatible packets, a single computed congruence modulo four is necessary and sufficient for a Leech lift. Complete proofs include a finite 24-row determinant certificate and an explicit kernel witness. The objects are classical; the contribution is the marked reconstruction and its exact integral boundary.

1. THE FIXED COORDINATE MODEL

Write $\Omega = \mathbb{F}_{23} \cup \{\infty\}$, serializing ∞ as coordinate 23. If Q is the set of nonzero squares modulo 23, let $\mathcal{C} \subset \mathbb{F}_2^{24}$ be the span of the 23 supports $(a + Q) \cup \{\infty\}$. A word is represented by the binary integer whose i th bit is its i th coordinate. The twelve words

85335e, 5606be, 2ca9ae, 11fe26, 0f55e2, 046f3e,
02379f, 0149a6, 00a4d3, 0042de, 003e4a, 001f25

form a basis; the notation is hexadecimal. Row reduction of the defining supports gives this basis. Enumerating its 2^{12} sums gives weights 0, 8, 12, 16, 24 with multiplicities 1, 759, 2576, 759, 1. This is the projective Golay code [Chapman(1997)]. We use the Leech model [Brouwer(n.d.), Theorem 1.4]

$$\Lambda = \left\{ x/\sqrt{8} : \begin{array}{l} x \in \mathbb{Z}^{24}, \quad x_i \equiv m \pmod{2} \text{ for one } m \in \{0, 1\}, \\ \sum_i x_i \equiv 4m \pmod{8}, \quad ((x_i - m)/2 \bmod 2)_i \in \mathcal{C} \end{array} \right\}. \quad (1)$$

Thus Λ is even, unimodular and has minimum squared norm four. These properties and the lattice itself are inherited, not claimed as new.

The three disjoint octads are

$$\begin{aligned} B_0 &= \{0, 1, 3, 4, 7, 8, 16, 23\}, \\ B_1 &= \{10, 12, 15, 18, 19, 20, 21, 22\}, \\ B_2 &= \{2, 5, 6, 9, 11, 13, 14, 17\}. \end{aligned}$$

Put $w_i = 2\mathbf{1}_{B_i}/\sqrt{8}$ and $P_i(x/\sqrt{8}) = \sum_{j \in B_i} x_j/4$. The matrices

$$C = \begin{pmatrix} 12 & 5 \\ 1 & 12 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 7 \\ 3 & 22 \end{pmatrix}, \quad T = \begin{pmatrix} 4 & 11 \\ 16 & 4 \end{pmatrix}$$

act on Ω by fractional linear transformations over $\mathbb{F}_{23}$ and permute lattice coordinates. They preserve $\mathcal{C}$; this can be checked on its twelve basis words. In projective action $T^4 = C$, and C cycles B_0, B_1, B_2 . All vector actions below are these coordinate permutations, not octonion-coordinate permutations.

2. THE SMALLEST STABLE LINEAR COMPLETION

For $n = 0, 1, 2, 3$ retain $P(T^n \lambda)$. Their sums agree, so they give nine independent integer readouts, denoted P_9 . Specifically, retain the first triple and the first two entries of each later triple.

Proposition 1. *The kernel of P_9 is not J -stable. The linear span of its nine functionals and their J -translates has dimension 15. The smallest $\langle J, T \rangle$ -stable real span containing them has dimension 24.*

Proof. The lattice vector with numerator

$$v = (8, -8, 8, 0, -8, 0, -8, 0, 8, 0, \dots, 0)$$

has $P_9(v/\sqrt{8}) = 0$ but

$$P_9(Jv/\sqrt{8}) = (0, 0, 0, -2, -2, 2, 0, 0, 2).$$

The code condition is trivial and the numerator sum is zero. Substitution verifies both readouts; rational row reduction gives ranks 9 and 15. For completeness of the stable span, act with J, T on B_0 , retaining

every distinct resulting support. This bounded calculation produces exactly the 759 weight-eight words of $\mathcal{C}$. The next theorem proves that their indicator functionals span rank 24. This proves minimality, not just the existence of one 24-dimensional completion. The enumeration rule is complete: begin with B_0 , apply both generators to every new support, and stop only when the list is closed. Both generators are invertible on the finite set. $\square$

Let $\mathcal{O}$ be the set of all 759 octads. For an arbitrary real vector x , put

$$a_B = \frac{1}{4} \sum_{i \in B} x_i, \quad S = \sum_{B \in \mathcal{O}} a_B, \quad S_i = \sum_{B \ni i} a_B.$$

Theorem 2 (Complete octad reconstruction). *The map $x \mapsto (a_B)_{B \in \mathcal{O}}$ is injective and has inverse on its image*

$$\boxed{x_i = \frac{23S_i - 7S}{1012}.} \quad (2)$$

Its real image consists exactly of data satisfying $a_B = \frac{1}{4} \sum_{i \in B} x_i(a)$ for all B . This is an explicit set of linear equations, not an existence condition.

Proof. Let A be the 759×24 indicator matrix. The Golay octads form the Steiner system $S(5, 8, 24)$ [Chapman(1997), Theorem 1]. Each coordinate lies in 253 octads and each pair in 77, by counting five-subsets. Hence

$$A^T A = 176I + 77\mathbf{1}\mathbf{1}^T.$$

In particular its rank is 24. Directly, $S = 253 \sum x_i/4$ and $S_i = 44x_i + 77 \sum x_j/4$; elimination yields (2). Applying the forward map tests precisely the displayed image equations. Both compositions are the identity on their stated domains. $\square$

3. EXACTLY ONE CYCLIC INTEGRAL OBSTRUCTION

Let $L_{\mathcal{O}}$ be the integral lattice generated by $2\mathbf{1}_B/\sqrt{8}$, for all $B \in \mathcal{O}$.

Lemma 3.

$$L_{\mathcal{O}} = \{x/\sqrt{8} \in \Lambda : \sum_i x_i \equiv 0 \pmod{16}\}, \quad [\Lambda : L_{\mathcal{O}}] = 4.$$

Moreover, with $v = \mathbf{1}/(2\sqrt{8})$,

$$L_{\mathcal{O}}^* = \Lambda + \mathbb{Z}v, \quad L_{\mathcal{O}}^*/\Lambda \cong \mathbb{Z}/4\mathbb{Z}.$$

Proof. Every generator satisfies the displayed congruence. On Λ the sum is divisible by four, and $x \mapsto \sum x_i/4 \pmod{4}$ is onto: the vector $(-3, 1, \dots, 1)/\sqrt{8}$ maps to 1. Thus the congruence sublattice has index four.

Here is a small, fully specified determinant certificate for the reverse inclusion. Sort the octad masks as increasing binary integers, numbered from zero. The 24 rows at positions

$$\begin{aligned} &197, 555, 724, 671, 582, 487, 580, 464, 333, 59, 49, 397, \\ &55, 427, 249, 304, 329, 72, 195, 630, 672, 16, 180, 402 \end{aligned}$$

have determinant 2^{14} . This is checked by exact fraction-free elimination. After scaling every row by $2/\sqrt{8}$, their covolume is $2^{14}/2^{12} = 4$. They generate a sublattice of the congruence sublattice with the same covolume, so equality holds.

Since Λ is integral, $\Lambda \subset L_{\mathcal{O}}^*$. Also $(v, 2\mathbf{1}_B/\sqrt{8}) = 1$, so $v \in L_{\mathcal{O}}^*$. The vector $4v$ lies in Λ , but $2v$ does not, by (1). The quotient $L_{\mathcal{O}}^*/\Lambda$ has order four by unimodularity and the index just proved. Hence v generates it. $\square$

Theorem 4 (Integral gluing syndrome). *Suppose $a \in \mathbb{Z}^{\mathcal{O}}$ satisfies the real image equations of Theorem 2. Then*

$$\delta(a) = \frac{11S - 23S_0}{506}$$

is an integer. There is a unique $\lambda = x/\sqrt{8} \in \Lambda$ having these octad traces if and only if

$$\boxed{\delta(a) \equiv 0 \pmod{4}.} \quad (3)$$

When it exists, (2) reconstructs all its coordinates. For example, the constant trace assignment $a_B = 1$ is linearly compatible but has $\delta = 5$, so it is not a Leech trace packet.

Proof. Compatibility gives a real vector pairing integrally with all generators of $L_{\mathcal{O}}$, hence an element of $L_{\mathcal{O}}^* = \Lambda + \mathbb{Z}v$. Set $v_0 = (-3, 1, \dots, 1)/\sqrt{8} \in \Lambda$. The homomorphism $z \mapsto 4(z, v_0) \bmod 4$ vanishes on Λ and sends v to $5 \bmod 4$, so it identifies that order-four quotient. In raw coordinates it is

$$4(x/\sqrt{8}, v_0) = (\sum_i x_i - 4x_0)/2 = (11S - 23S_0)/506.$$

This proves integrality, the criterion, and uniqueness. Constant traces reconstruct $x_i = 1/2$, that is v ; their syndrome is five. $\square$

4. SCOPE AND VERIFICATION

The output is a complete integral coordinate theorem and a minimal linear symmetry completion. It does not assert that the additional Erdős–Straus cubic contains a positive integral trace. The data and maps are classical-code constructions; no new lattice, uniqueness theorem, or historical-priority claim is made. A content-level search of Golay design lattices, Smith invariants, octad trios and octonionic models did not establish a prior exact statement of this combined marked reconstruction; it also cannot establish absence from the literature.

The standalone `verify_audit.py` reconstructs the code, the generator orbit, all entries of $A^T A$, the determinant minor, the kernel witness, and exact inverse/syndrome tests. The finite facts entering the proof have fixed bounds (4096 codewords, 759 octads, a 24×24 determinant). No floating point, unbounded search, formal-prover run or independent reviewer is used. The general assertions follow from the arguments above, not from extrapolating these tests.

REFERENCES

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